

# BioSpecInfo

## Software Bill of Materials (SBOM)

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Versione **bsi-v176** · commit `ce18622`

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Documento generato automaticamente dal sorgente Markdown versionato nel repository. La versione e il commit stampati in ogni piè di pagina identificano esattamente lo stato del software descritto.

# Software Bill of Materials (SBOM)

BioSpecInfo · document generated by `tools/genera-sbom.js`.

|                         |                                                        |
|-------------------------|--------------------------------------------------------|
| Application version     | bsi-v176                                               |
| Commit                  | 57a40bd02a670269df36b3ca5b70b05f4402d0a9               |
| Generated (UTC)         | 2026-10-01T23:25:19.944Z                               |
| Machine-readable format | <code>../evidence/sbom.cdx.json</code> — CycloneDX 1.5 |

The SHA-256 digests refer to the files actually distributed in this commit. A reviewer can recompute them with `sha256sum <file>` and compare, confirming they hold the very artefact described here.

## 1. Third-party components distributed with the application

Every component is **included in the repository** and served from the same origin: the application loads no code from external CDNs at runtime. That is what makes it usable offline and what removes any dependence on third-party availability.

| Component                                                                 | Version/File                      | Licence      | Size    | Role                                                                 |
|---------------------------------------------------------------------------|-----------------------------------|--------------|---------|----------------------------------------------------------------------|
| <b>RDKit MinimalLib (JS glue)</b> <br><sub>RDKit / Greg Landrum</sub>     | <code>RDKit_minimal.js</code>     | BSD-3-Clause | 125 kB  | Cheminformatics: SMILES and SMARTS parsing, descriptors, 2D drawing. |
| <b>RDKit MinimalLib (WebAssembly)</b> <br><sub>RDKit / Greg Landrum</sub> | <code>RDKit_minimal.wasm</code>   | BSD-3-Clause | 6753 kB | WebAssembly binary of the cheminformatics engine.                    |
| <b>three.js</b> <br><sub>mrdoob e collaboratori</sub>                     | <code>three.min.js</code>         | MIT          | 589 kB  | WebGL rendering engine for the 3D scenes.                            |
| <b>three.js — UnrealBloomPass</b> <br><sub>mrdoob e collaboratori</sub>   | <code>three_bloom.js</code>       | MIT          | 25 kB   | Post-processing effect for the astronomical visualisations.          |
| <b>three.js — GLTFLoader</b> <br><sub>mrdoob e collaboratori</sub>        | <code>gltf_loader.js</code>       | MIT          | 94 kB   | Loading of anatomical and molecular models in glTF format.           |
| <b>3Dmol.js</b> <br><sub>University of Pittsburgh / David Koes</sub>      | <code>3Dmol-min.js</code>         | BSD-3-Clause | 502 kB  | 3D molecular visualisation (PDB, SDF, surfaces).                     |
| <b>SmilesDrawer</b> <br><sub>Daniel Probst</sub>                          | <code>smiles-drawer.min.js</code> | MIT          | 246 kB  | 2D structure drawing from SMILES.                                    |
| <b>sql.js (JS glue)</b> <br><sub>sql.js / SQLite</sub>                    | <code>lib/sql-wasm.js</code>      | MIT          | 45 kB   | In-browser SQLite database for the data exercises.                   |

| Component                                                                    | Version/File      | Licence       | Size   | Role                                                 |
|------------------------------------------------------------------------------|-------------------|---------------|--------|------------------------------------------------------|
| <b>SQLite (WebAssembly)</b> <br><sub>SQLite Consortium</sub>                 | lib/sql-wasm.wasm | Public Domain | 644 kB | SQLite engine compiled to WebAssembly.               |
| <b>Dataset dimuoni (CERN Open Data)</b> <br><sub>CERN Open Data Portal</sub> | lib/dimuon.js     | CC0-1.0       | 48 kB  | Experimental data for the particle-physics exercise. |

## Digests

| File                 | SHA-256                                                          |
|----------------------|------------------------------------------------------------------|
| RDKit_minimal.js     | 58d3c996ade7e0b0137d4f9363ff6c204b545689428fa0898eeaed579ef788d9 |
| RDKit_minimal.wasm   | e0967d44fed59e44a2d07bfc08f3a86821c54a14c95bb7e6e392fe867333b8c8 |
| three.min.js         | 9274bbcec8d96168626c732b5d31c775aa8cfb7eaa0599bec0c175908a2c1ce2 |
| three_bloom.js       | 42e0ff40b00ea1ba4fe53baeb3726834a3ac3db46e9cfb80856baebf01611090 |
| gltf_loader.js       | 5c15967ba830918a9caea6338712c994c354bccd4edc4569bde411c3ec06a3e6 |
| 3Dmol-min.js         | 06b6d2fc7d418e8bef62a32cca44373ff21b2b787bd377613dd4039394a4e9ff |
| smiles-drawer.min.js | 955285fac52a7017b3c3658e2cb84bef7f1f76732f3f7cd565d1bb0903232dc2 |
| lib/sql-wasm.js      | 77d6435bac506af0e3c59636dce9d22b1b14156348bc327f41a1577f3212360f |
| lib/sql-wasm.wasm    | 438c88f666dc054ce4e9395f80fe9db4218b1a3c379960454880f048a7898aed |
| lib/dimuon.js        | 61190d7bbec3ce4ea3add2503b9513ac27b227a3878dd229560a6cd29aabc36d |

## 2. Code written for this project

| File                   | Size    | Role                                                                                                   |
|------------------------|---------|--------------------------------------------------------------------------------------------------------|
| index.html             | 4576 kB | Main application: interface, teaching sections, chemical data.                                         |
| bsi-ai-hub.js          | 422 kB  | «Spectra» AI agent: agentic loop, tools, provider management.                                          |
| bsi-spettri.js         | 32 kB   | IR/NMR spectral prediction engine over the molecular graph.                                            |
| bsi-cheminfo.js        | 65 kB   | Cheminformatics engine: standardisation, fingerprints, clustering, PCA, QSAR models with a null model. |
| sw.js                  | 5 kB    | Service Worker: offline operation and network strategy.                                                |
| rdkit_lab.html         | 260 kB  | Cheminformatics laboratory.                                                                            |
| astro.html             | 2666 kB | Astrochemistry module.                                                                                 |
| chimorga.html          | 194 kB  | Organic chemistry module.                                                                              |
| proxy/spectra-proxy.js | 10 kB   | Optional proxy (Cloudflare Worker) for the AI calls.                                                   |

## Digests

| File                   | SHA-256                                                          |
|------------------------|------------------------------------------------------------------|
| index.html             | 2fa8276caac616aa5f5a4ce0969c52e848126a9ddc8dc05786a2f25ec0eb2e94 |
| bsi-ai-hub.js          | 985b9fa3f6a3d330cc6f7e31456e16760d0a0d9482f760b04d83765a013f2476 |
| bsi-spettri.js         | 6f6d13824b593c92edfdf51cd976de4cb72479c9c6586c5158c56dab615a779b |
| bsi-cheminfo.js        | aefd53302644ef47f4cd78a6f531a8353e4952d9d4fc1eaf006a2f490e7fca9f |
| sw.js                  | bef94eeb13a8350937ba6af043bce0c874333a3e492582d0a7ffc4368c3d3b6f |
| rdkit_lab.html         | 066a42092b829df2fa306b691b9d162b6d44edfdb5317ce4cc8a276bc34c80ad |
| astro.html             | 53e8d4d29af4bb9cfdb63d974924737e606e3141bd4ce61fbab1bbebe0eed30d |
| chimorga.html          | 047934abc73a79fbaf06ed7036b6e89a913ebdf5354fc52cecdbee7e8f94f51d |
| proxy/spectra-proxy.js | 808c0a12891ae9c6f45cf0c676b033075089e064df9047be844f03badf940962 |

## 3. Development dependencies

Used **only** to run the verification benches. They are not distributed with the application and are not present in production.

| Package         | Version | Licence    |
|-----------------|---------|------------|
| playwright-core | 1.62.1  | Apache-2.0 |

## 4. Runtime dependencies

**None.** The application has no backend, installs no package at runtime and loads no script from external domains. The only network calls are optional and go to documented public services:

| Service                      | Purpose                                        | If unreachable                                   |
|------------------------------|------------------------------------------------|--------------------------------------------------|
| PubChem PUG-REST             | IUPAC name, CAS, GHS pictograms, 3D conformers | the app uses local data and says so              |
| NASA / ESA                   | astronomical images and data                   | the module shows the tabulated data              |
| AI providers (user's choice) | assistant answers                              | the assistant reports it and offers alternatives |

No personal data is transmitted: API keys, if entered, stay in the device's `localStorage`, or on the user's own proxy (see `proxy/`).

## 5. Licence compatibility

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All licences of the distributed components (MIT, BSD-3-Clause, CC0, Public Domain) are **permissive** and mutually compatible. No component is subject to a strong copyleft licence (GPL/AGPL): there is no redistribution constraint on the code written for this project.

The remaining obligations are of **attribution**: the licence notices of the upstream projects are reproduced in `THIRD_PARTY_NOTICES.md`.

**The application's licence is distinct from those of its components.** The code written for BioSpecInfo is **proprietary** — *All rights reserved* (see [LICENSE](#)): readable and open to assessment, but copying, reuse and commercial use require the Author's written permission. The permissive licences listed above cover **only** the bundled third-party libraries, and do not extend to the application.

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